

KUMARAGURU
Institutions

2026

**Micro
cosm**
The Co-existing Architecture

Carbon Emissions Report 2025

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Introduction

Kumaraguru Institutions is a multi-campus educational institution in Tamil Nadu, South India, offering undergraduate, postgraduate, and research programs in Engineering, Technology, Management, Agriculture, Liberal Arts, and Science. Guided by its motto, "Character is Life," the institution is committed to academic excellence, holistic development, and nurturing responsible individuals with integrity and purpose. Through experiential learning, innovative research, and strong community engagement, Kumaraguru fosters a culture of inquiry and social responsibility, empowering students to address real-world challenges and contribute meaningfully to society.



Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance, resource efficiency, and responsible environmental stewardship.

This document follows Greenhouse Gas Protocol and ISO 14064 standards to provide a transparent account of our Greenhouse Gas (GHG) inventory across energy, transport, and waste. By quantifying these operational impacts, we establish a robust baseline for targeted reduction strategies and reaffirm our continuous commitment to environmental sustainability.

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Microcosm Logo

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Highlights



EMISSION PROFILE

Total Gross GHG

277.39 tCO₂e

Scope 1
(DG +
Transport)

122.7 tCO₂e

Scope 2
(Electricity)

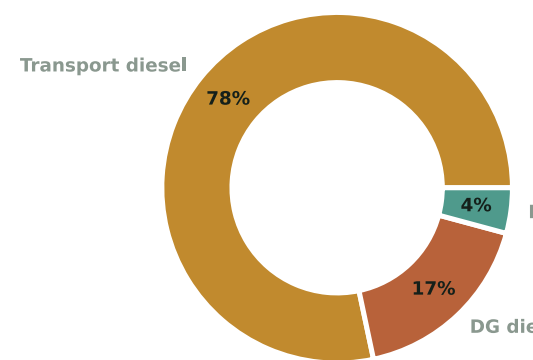
154.7 tCO₂e

Scope 3
(Others/
Offset)

0.0 tCO₂e

Emission
Contribution

Scope 1 Breakdown



ENERGY INDICATORS

Total Energy

537566 kWh

Grid
Electricity

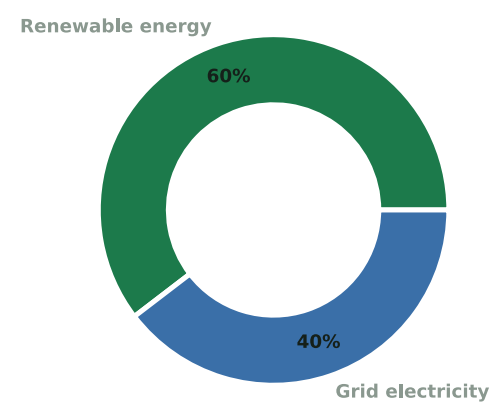
212822 kWh

Renewable
Energy

324744 kWh

Energy Split

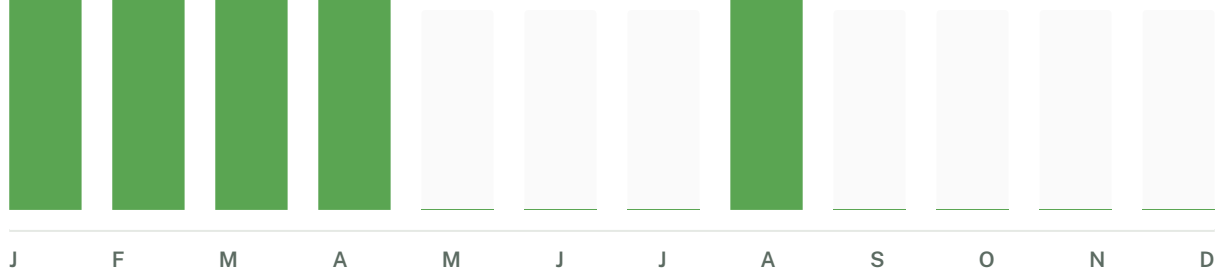
Renewable Mix



Renewable
Share

60.4 %

Monthly Grid Energy Trend (kWh)



Transport Indicators



FLEET & FUEL OVERVIEW

Total Fleet Fuel Consumption

37729 Litres

Fleet Carbon Footprint

101.2 tCO₂e

Petrol vs Diesel (Litres)

2154.0 L

Petrol (5.7%)

35574.6 L

Diesel (94.3%)



Monthly Transport Fuel
(Petrol+Diesel)

Petrol vs Diesel (Emissions)

5.1 tCO₂e

Petrol (5.1%)

96.1 tCO₂e

Diesel (94.9%)



J F M A M J J A S O N D

Transport Fleet Insights

Transport emissions increased 11.7%. Primary cause: Diesel fleet usage.
Suggested improvement: Electrify campus buses.

Climate Action & Mitigation



SUSTAINABILITY INITIATIVES

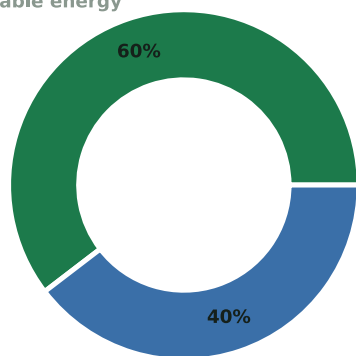
Total Avoided Emissions via Renewables

236.1 tCO₂e

Renewable Integration Impact

Renewable Mix

Renewable energy



Grid electricity

Per Capita Intensity

0.000 tCO₂e / person

Net Zero Progress Target (Illustrative)

236.1 tCO₂e

Avoided (46.0%)

277.4 tCO₂e

Remaining (54.0%)



Decarbonization Story

Renewable energy integration contributed 324744.00 kWh, offsetting approximately 236.09 tCO₂e of potential emissions. Our current renewable share stands at 60.4%.

Executive Summary

High-level overview of carbon performance for Carbon Emissions Report - 2026.

Performance Overview

During this reporting period, the campus generated 277.39 tCO₂e of gross carbon emissions. Through our renewable energy initiatives, we successfully avoided 236.09 tCO₂e, resulting in a net carbon impact of 41.30 tCO₂e. We remain committed to our long-term decarbonization pathway.



CARBON STORY

Gross GHG Emissions

277.4 tCO₂e

Net GHG Emissions

41.3 tCO₂e

Avoided Emissions

236.1 tCO₂e



STRATEGIC RECOMMENDATIONS

- Accelerate the transition of internal campus mobility to Electric Vehicles (EVs) to reduce Scope 1 transport emissions.
- Expand rooftop solar capacity to push the renewable energy mix beyond the current baseline.
- Implement automated BMS (Building Management Systems) for high-load commercial blocks to optimize HVAC consumption.

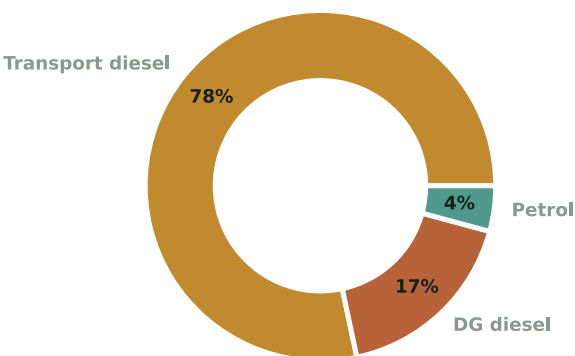
Scope 1 Story

Direct Emissions Overview

Scope 1 emissions generated 122.67 tCO₂e. Transport remains the dominant emission source in this category. Overall Scope 1 emissions increased by 19.4% compared to the previous period.

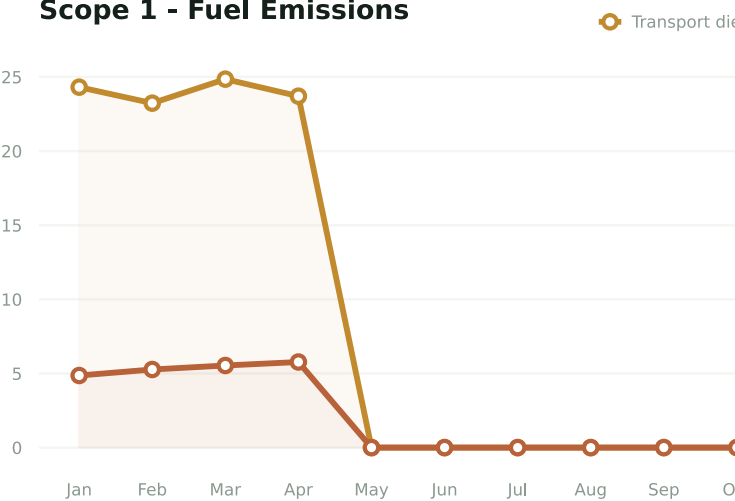
Scope 1 Composition

Scope 1 Breakdown



Emissions Trend

Scope 1 - Fuel Emissions



Transport Deep Dive

Transport emissions increased 11.7%. Primary cause: Diesel fleet usage. Suggested improvement: Electrify campus buses.

Transport Emissions (Scope 1)

101.2 tCO₂e

MONTH	PETROL (LITRES)	EMISSIONS (TCO ₂ E)	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	585.7	1.3986	9000	24.3090
FEB	520.0	1.2418	8600	23.2286
MAR	540.0	1.2895	9200	24.8492
APR	508.4	1.2140	8775	23.7002
MAY	0.0	0.0000	0	0.0000
JUN	0.0	0.0000	0	0.0000
JUL	0.0	0.0000	0	0.0000
AUG	0.0	0.0000	0	0.0000
SEP	0.0	0.0000	0	0.0000
OCT	0.0	0.0000	0	0.0000
NOV	0.0	0.0000	0	0.0000
DEC	0.0	0.0000	0	0.0000

Transport Insights

The table above highlights the monthly breakdown of combustion fuels used in institutional mobility. Diesel remains the primary fuel source for heavy transport and buses.

Cumulative Fuel Mix

2154.0 L
Petrol (5.7%)

35574.6 L
Diesel (94.3%)



DG Set Emissions (Scope 1)

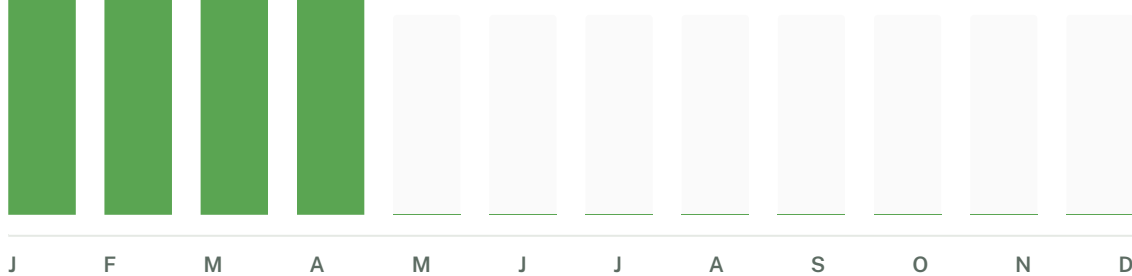
21.4 tCO₂e

MONTH	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	1800.00	4.8618
FEB	1950.00	5.2669
MAR	2050.00	5.5370
APR	2136.83	5.7716
MAY	0.00	0.0000
JUN	0.00	0.0000
JUL	0.00	0.0000
AUG	0.00	0.0000
SEP	0.00	0.0000
OCT	0.00	0.0000
NOV	0.00	0.0000
DEC	0.00	0.0000

DG Operations Analysis

Diesel Generator usage is strictly reserved for grid outages and critical power backups. Optimizing backup load management helps minimize these localized emissions.

DG Monthly Trend



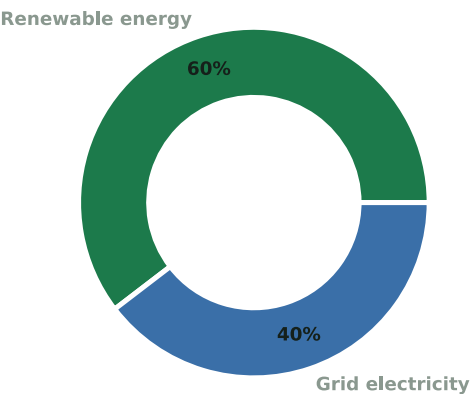
Scope 2 Story

Indirect Emissions Overview

Grid electricity consumption accounted for 154.72 tCO₂e. This represents 55.8% of total gross emissions. Scope 2 emissions fell by 71.8%. Continuous efforts in optimizing HVAC and lighting can further reduce this footprint.

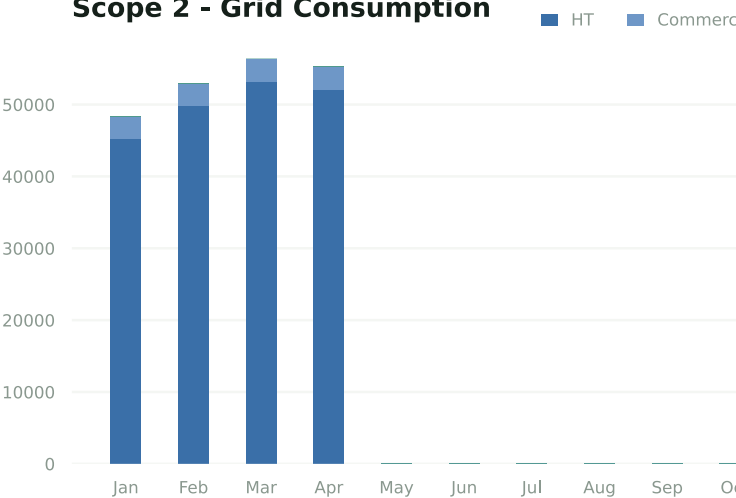
Renewable Energy Mix

Renewable Mix



Grid Consumption Profile

Scope 2 - Grid Consumption



Renewable Impact

Renewable energy integration contributed 324744.00 kWh, offsetting approximately 236.09 tCO₂e of potential emissions. Our current renewable share stands at 60.4%.

Electricity Emissions (Scope 2)

154.7 tCO2e

RENEWABLE ENERGY

324744 kWh

GRID ENERGY

212822 kWh

TOTAL CONSUMPTION

537566 kWh

MONTH	GRID HT		GRID COMMERCIAL		GRID TEMPORARY	
	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)
JAN	45208	32.866	3100	2.254	0	0.000
FEB	49822	36.221	3042	2.212	0	0.000
MAR	53200	38.676	3200	2.326	0	0.000
APR	52065	37.851	3185	2.315	0	0.000
MAY	0	0.000	0	0.000	0	0.000
JUN	0	0.000	0	0.000	0	0.000
JUL	0	0.000	0	0.000	0	0.000
AUG	100	0.073	0	0.000	0	0.000
SEP	0	0.000	0	0.000	0	0.000
OCT	0	0.000	0	0.000	0	0.000

MONTH	GRID HT		GRID COMMERCIAL		GRID TEMPORARY	
	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)
NOV	0	0.000	0	0.000	0	0.000
DEC	0	0.000	0	0.000	0	0.000

Grid Operations Insights

The detailed monthly tracking indicates peak grid consumption variations. Industrial High Tension (HT) lines remain the largest contributor to our Scope 2 profile.

Grid vs Renewable Generation



Emissions Comparison

Performance analysis vs previous reporting period



CONSOLIDATED GHG SUMMARY

CATEGORY	CURRENT PERIOD	PREVIOUS PERIOD	% CHANGE
Total Emissions	277.39	651.11	-57.40%
Scope 1 Emissions	122.67	102.70	+19.44%
Scope 2 Emissions	154.72	548.41	-71.79%
Renewable Energy (kWh)	324744	1124456	-71.12%

Current vs Previous Gross Emissions

277.4 tCO₂e

Current (29.9%)

651.1 tCO₂e

Previous (70.1%)



Progress Summary

The comparison table above illustrates our performance trajectory against the previous baseline. Reductions in total emissions are primarily driven by continuous optimization in energy usage and sustained renewable integration.

Methodology & Conclusion

Reporting Standards and Final Insights

Purpose of the Report

This report presents the Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions for the reporting period, compiled in accordance with ISO 14064-1:2018. The objective of this inventory is to measure and disclose the institution's GHG emissions, and provide reliable data to support future emissions reduction strategies.

GHG Quantification Approach

The inventory has been developed using an activity-based quantification approach. Emissions were calculated by multiplying activity data (e.g., fuel consumption, electricity consumption) with appropriate emission factors derived from recognized national and international sources (IPCC, CEA, DEFRA).

Organizational Boundary

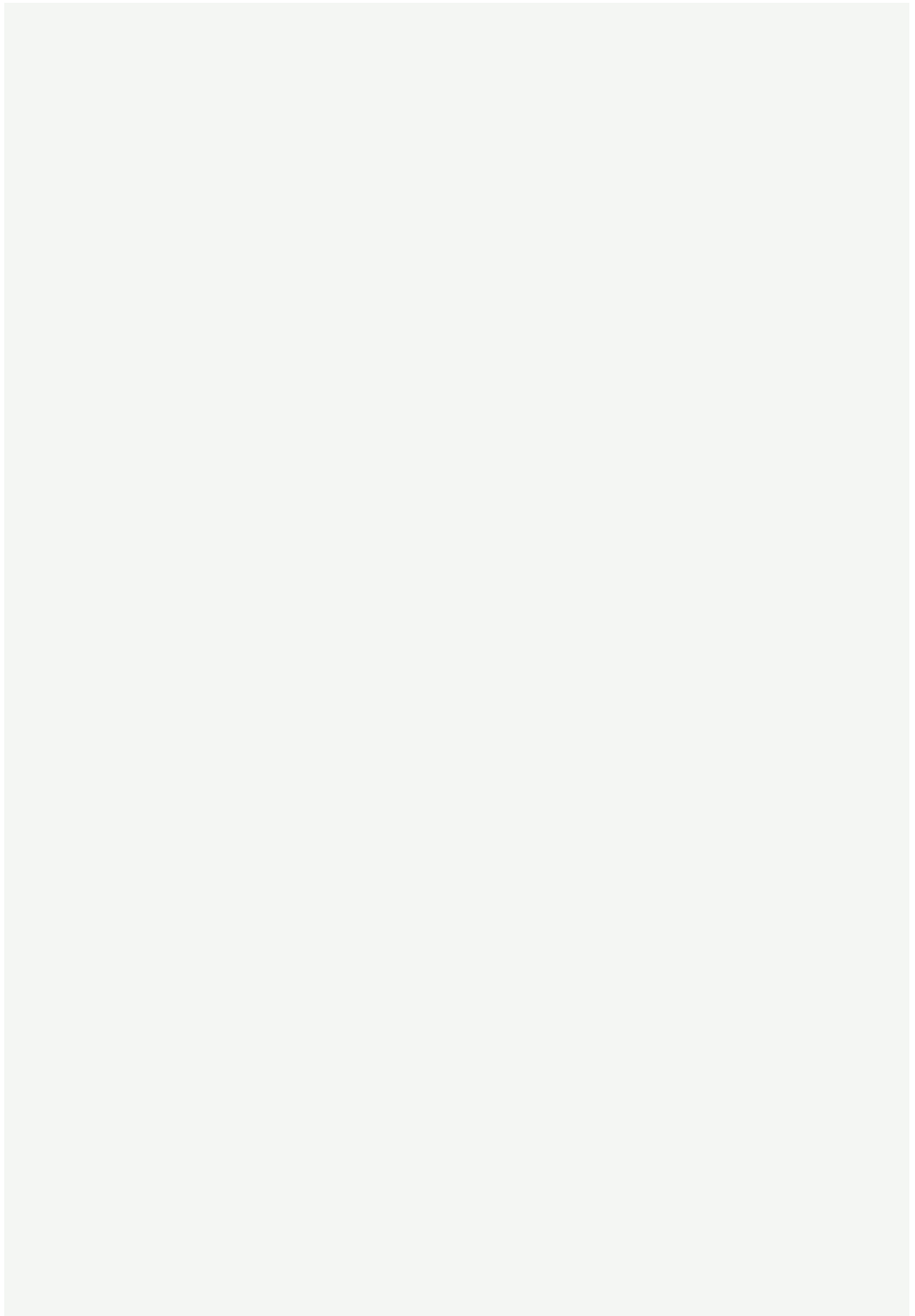
The organizational boundary has been defined using the operational control approach. All facilities and operations over which Kumaraguru Institutions has operational control are included. Scope 3 emissions have been assessed at a screening level but are excluded from quantitative reporting due to the unavailability of consolidated data.

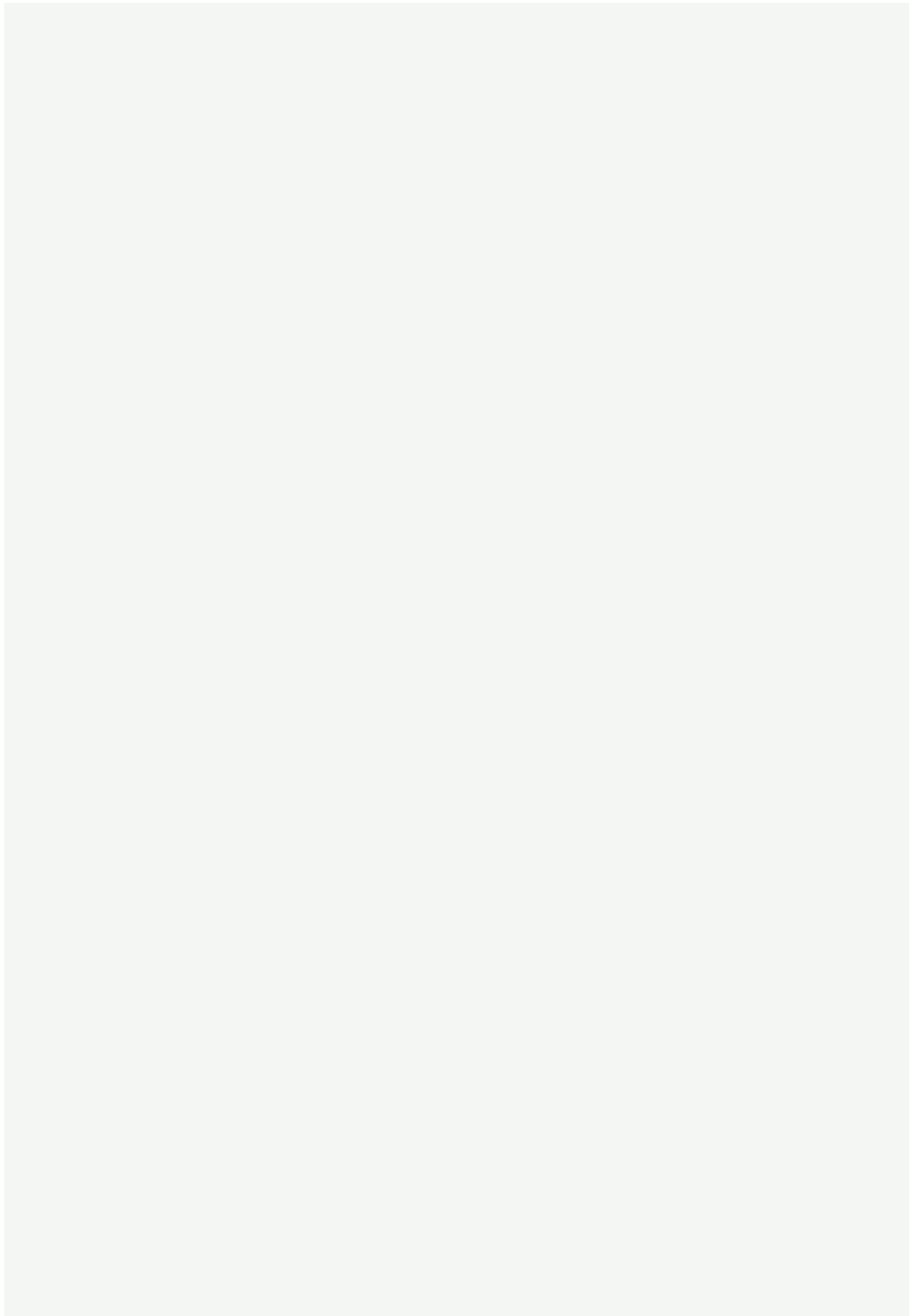
Standard Emission Factors Applied

EMISSION SOURCE	FACTOR	UNIT
Grid Electricity (TN)	0.727	kgCO ₂ e / kWh
Diesel Fuel	2.701	kgCO ₂ e / L
Petrol Fuel	2.388	kgCO ₂ e / L

Conclusion

This GHG inventory enhances the understanding of emission sources within campus operations and provides a foundation for tracking environmental performance. The institution intends to progressively improve data quality, expand emission coverage, and integrate climate considerations into operational decision-making.





REPORT PREPARED BY

Microcosm Sustainability Intelligence

A long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance.